

Problem Statement:

Employee attrition poses a significant challenge to any organization as it creates adverse effects on production, increases expenses due to the need for replacements, and overall adversely affects business processes. This paper aims to examine the patterns in the data provided in the HR Employee Attrition Set in order to determine factors that contribute to employee attrition. The analysis will enable drawing conclusions on the way to maintain employees and ensure their satisfaction within the workplace.

Questions:

1. What percentage of employees leave the organization?
2. Which departments have the highest number of employees?
3. What is the age distribution of employees?
4. How is monthly income distributed among employees?
5. What are the job satisfaction levels of employees?
6. Which employee characteristics might contribute to attrition?

Analysis Plan:

To address the problem statement, the following analytical approach will be used:

1. Data Collection

- Use the publicly available HR Employee Attrition dataset.

2. Data Exploration

- Understand the dataset structure.

- Identify missing and duplicate values.
- Review data types and summary statistics.

3. Exploratory Data Analysis (EDA)

- Visualize employee attrition.
- Analyze employee age distribution.
- Examine department-wise employee distribution.
- Study monthly income distribution.
- Explore job satisfaction levels.

4. Tools and Technologies

- Python
- Pandas
- Matplotlib
- Seaborn
- Google Colab